

Concept Research Dossier

Three independent adversarial analyses: consumer concepts, merchant concepts, and the winning-mechanism selection.

Source of truth: *Paytm_AI_Hackathon_Themes_Hyderabad.pdf* (2 pages, 2 themes). Judging model assumed: Paytm Integration, AI Innovation, User Impact, Demo Quality, Build Feasibility — 20 points each, 100 total. All scores are analyst estimates, not official marks.

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2. **Part II — Theme 2: AI for Small Businesses.** Eight merchant concepts across growth, money, inventory, billing and planning.
3. **Part III — Winning mechanism.** Cross-theme selection, competitor field analysis, red-team review, and a deterministic three-minute demo.
4. **Part IV — Reconciling the three analyses.** Where they disagree and what that implies.

The binding constraint from the PDF. Each theme defines a problem space, not a solution. Every solution “must be built for the existing Paytm stack” and must “plug into Paytm's current products and ecosystem rather than stand alone as an unrelated idea.” Theme 1 solutions must work “as a feature within the current Paytm user experience, not as a separate product.” Theme 2 solutions must “extend the products a Paytm merchant already uses.” This is the criterion most teams will fail.

Part I — Theme 1: AI for Paytm Users

Verdict up front: the two worth funding are **Rakshak** (pre-payment scam interstitial) and **Ek Photo Setup** (camera-first bill-stack onboarding).

Three ideas were discarded immediately, because roughly half of any Theme 1 field submits them: a Paytm chat assistant, “ask your transaction history a question,” and a voice-controlled navigation layer. They fail the same test — the AI is a front door to features that already exist, so AI innovation collapses to prompt engineering and user impact is a convenience delta rather than a behaviour change. The analysis weights heavily toward ideas where Paytm's *proprietary data* is the moat, not just Paytm's *screens*.

Scoring calibration. 20 means a Paytm PM would put this on the roadmap Monday. 15 is solid. 12 is “works, but a judge will poke a hole in it.” Nothing earns 20 on feasibility in a one-day sprint.

1. Rakshak — the pre-payment scam interstitial

MECHANISM

Between Confirm and Pay, a full-screen card appears for anomalous payees: “This UPI ID first appeared 31 hours ago. It has received money from 340 people across 11 states today. 22 of them have filed complaints.” Below it, one line in the user's language explaining the *specific* scam pattern matched, with a hard “Hold, don't pay” as the primary button and “Pay anyway” demoted.

WHY AI IS NECESSARY

Two models doing real work: a graph-anomaly model over the payee's inbound edge structure (velocity, geographic fan-in, first-timer ratio, device churn), plus a small NLP model classifying the *pretext* — if the user pasted a VPA from a message, or the note says “electricity bill disconnection,” it matches a known fraud script. Rules alone cannot do this; mule accounts mutate faster than a rules committee ships.

WHY PAYTM IS NECESSARY

Total. Unbuildable by anyone without a national UPI transaction graph. A competitor cannot copy it and no standalone startup can build it. The most defensible idea in the set.

META FIT

Genuinely natural: most of these scams *arrive* on WhatsApp. A “forward the suspicious message to Paytm” number that returns a verdict in the user's language extends the shield from the payment moment to the persuasion moment.

WOW MOMENT

A judge scans a QR pre-seeded as a mule account. The screen goes amber and says, in Tamil, “Nine people were cheated by this shop code today.” The room goes quiet.

FEASIBILITY

UI and inference path are half a day. The problem is data — real mule graphs will not be available. Ship a synthetic graph of roughly 50k edges and be openly honest on stage that it is synthetic while the feature schema matches what Paytm actually logs.

BIGGEST FLAW

False positives are catastrophic in a payments app. A legitimate new merchant flagged as a fraudster is a merchant-relations incident, and any judge with risk experience will ask about the precision/recall trade-off within thirty seconds. The answer must be ready: this fires on the top 0.3% of risk scores, not on everything unusual.

Paytm 20 · AI 16 · Impact 19 · Demo 17 · Feasibility 13 → 85

2. Bol-ke-Bhej — voice payment with vernacular read-back

MECHANISM

Hold the mic and say “Ramesh bhaiya ko do hazaar bhejo” in Bhojpuri, Marathi, whatever. The AI resolves the payee from contacts and payment history, then — the actual innovation — the confirmation screen *speaks back* in the same language: the amount in words, the payee's real registered name, and their photo. Two-way voice, not one-way command.

WHY AI IS NECESSARY

Real but unglamorous. Low-resource Indian-language ASR with code-mixing (“do hazaar” inside an English UI) is where off-the-shelf models break, and payee disambiguation across three contacts named Ramesh needs history-based ranking.

WHY PAYTM IS NECESSARY

Moderate. The contact graph and payment history make it good, but a competitor could ship the same thing. This is the weak leg.

META FIT

Llama-based on-device inference is defensible for languages with no cloud API coverage, and WhatsApp voice notes are already the input habit for this exact user.

WOW MOMENT

A judge speaks in their mother tongue and money moves. The most viscerally impressive demo in the set — and the read-back doubles as a fraud check, since scam victims typically never verify the payee's registered name.

FEASIBILITY

High, and the safest build. Caveat: live microphone demos fail in noisy rooms. Pre-record a fallback.

BIGGEST FLAW

A judge will say “Google Pay has voice, and so does every assistant.” The entire defence rests on read-back confirmation being a *trust* feature rather than a *speed* feature. Pitched as speed, it loses.

Paytm 17 · AI 13 · Impact 18 · Demo 18 · Feasibility 16 → 82

3. Ek Photo Setup — camera-first onboarding that builds your bill stack

MECHANISM

A brand-new user's first screen is not a form, it is a camera. Photograph any electricity bill, gas receipt or broadband invoice. The AI extracts biller identity, consumer number, billing cycle and typical amount, matches against the BBPS biller catalogue, and configures the account. Three documents and the user's entire recurring-payments life is set up with due-date reminders — zero typing.

WHY AI IS NECESSARY

Genuine document understanding across hundreds of state utility formats, most badly printed, many bilingual, none standardised. Consumer-number extraction is the hard part: a differently-labelled, differently-positioned field on every biller's template. Templated OCR has been tried and fails at scale; this is exactly what vision-language models unlocked.

WHY PAYTM IS NECESSARY

Very high. Without the BBPS biller catalogue and payment rails, extracting a consumer number is a party trick. The extraction has value only because Paytm can immediately *act* on it.

META FIT

Natural — utility bills increasingly arrive as WhatsApp PDFs. “Forward your bill to Paytm on WhatsApp, we'll set it up” removes even the photography step.

WOW MOMENT

Fresh install, camera, one photo of a crumpled electricity bill, and four seconds later the home screen has a configured biller with the correct due date. Onboarding friction for first-time users is named in the brief, and this attacks it at the root rather than smoothing it.

FEASIBILITY

Strong. One vision-model call, a fuzzy match against a hardcoded biller list for the demo, and a confirmation screen. Two people can build it in a day.

BIGGEST FLAW

It is a one-time-use feature. A judge will ask about retention: day zero is solved brilliantly and then the feature never fires again. The counter is that day zero is where Paytm loses first-time users, so a one-time feature at the highest-drop-off point is worth more than a daily feature at a low-drop-off point.

Paytm 19 · AI 15 · Impact 16 · Demo 19 · Feasibility 15 → 84

4. Paisa Kahan Gaya — merchant-name decoder and mandate killer

MECHANISM

Statements are unreadable: `PAYTM*RZP*QWK9812*MUM`. The AI normalises every line into a real merchant with a real logo, clusters recurring charges, and surfaces the ones silently draining money — the trial that converted, the app deleted six months ago that still has a live UPI mandate. One tap revokes the mandate.

WHY AI IS NECESSARY

Moderate but real. Merchant-string normalisation is a genuinely nasty entity-resolution problem nobody solves well, and recurrence detection over irregular intervals needs a model rather than a cron job.

WHY PAYTM IS NECESSARY

High. The UPI Autopay mandate registry is the asset, and the ability to *revoke* in one tap — not merely inform — is what makes this a Paytm feature rather than a spreadsheet.

META FIT

None natural. Do not force one; judges notice.

WOW MOMENT

“You are paying ₹2,847 a month for eleven subscriptions. You have opened four of them this year.” Then a single tap kills three. The number does the work.

FEASIBILITY

High — mostly data plumbing over a synthetic statement, no exotic models.

BIGGEST FLAW

Every neobank in India has shipped a version of this, and the AI content is thin enough that AI innovation scores poorly. It is also, awkwardly, a feature that reduces payment volume on your own platform.

Paytm 18 · AI 14 · Impact 15 · Demo 13 · Feasibility 17 → 77

5. Parchi — handwritten kirana bill to itemised payment

MECHANISM

Photograph the handwritten slip the kirana shop hands over. The AI reads the Devanagari scrawl, itemises it, checks the arithmetic (shopkeepers make errors, sometimes not accidentally), flags discrepancies, and pays the total via the shop's QR — producing an itemised digital record neither party had before.

WHY AI IS NECESSARY

Extremely high. Handwritten mixed-script recognition with numerals in the margins is close to the hardest thing on this list, which is exactly why it scores well on innovation.

WHY PAYTM IS NECESSARY

Moderate. The QR merchant network makes payment seamless, but the OCR itself is portable. A judge will notice the AI is doing the impressive part while Paytm does the easy part.

META FIT

Weak to moderate; shop orders do travel over WhatsApp images, but it is not load-bearing.

WOW MOMENT

Undeniably strong when it works — a scribbled paper slip becomes a clean itemised bill with “₹12 addition error” highlighted.

FEASIBILITY

This is the trap. Handwriting accuracy on real kirana slips is nowhere near demo-safe, and the failure mode is public and embarrassing. Misreading ₹120 as ₹1,200 on stage ends the run.

BIGGEST FLAW

Accuracy risk in a domain where a single-digit error means real money lost. Also mildly adversarial toward the merchants Paytm depends on — “our app catches your shopkeeper cheating” is not a partnership-friendly narrative.

Paytm 15 · AI 17 · Impact 14 · Demo 18 · Feasibility 12 → 76

6. Dispute Autopilot — failed-transaction resolution agent

MECHANISM

For a UPI failure where money debited but never arrived, instead of a support queue the AI reads the failure code and rail state, decides whether it is reversal-pending or a genuine dispute, drafts and files the chargeback with the correct bank, and reports plain-language status in the user's language with an honest ETA.

WHY AI IS NECESSARY

Weak, honestly. Roughly 80% deterministic workflow and 20% language generation. A judge will correctly identify it as automation wearing an AI hat.

WHY PAYTM IS NECESSARY

Very high — impossible without transaction records and PSP relationships.

META FIT

WhatsApp status updates are natural, since the user should not have to reopen the app to check.

WOW MOMENT

Weak. There is no visual one; the payoff is asynchronous and unfolds over days.

FEASIBILITY

Poor for a hackathon. The entire value sits in inaccessible backend integrations, so the demo is necessarily a mock — and a mocked backend with no visible AI is the worst possible combination.

BIGGEST FLAW

Highest real-world user impact on the list and the worst hackathon fit. A Q3 roadmap item, not a one-day build. Abandon it despite believing in it.

Paytm 19 · AI 12 · Impact 18 · Demo 11 · Feasibility 12 → 72

7. Nakad Nazar — cash-flow runway for irregular incomes

MECHANISM

For gig workers and daily-wage earners with lumpy inflows, the AI forecasts from UPI patterns: “Based on your last four months, you'll be about ₹800 short around the 27th, when your rent and two bills land together.” It then suggests moving one flexible bill by four days rather than pushing credit.

WHY AI IS NECESSARY

High. Forecasting sparse, irregular, seasonally-noisy income is a real time-series problem, and the population that needs it most is the one traditional credit models describe worst.

WHY PAYTM IS NECESSARY

High. Only Paytm sees both sides — merchant inflows and bill outflows — for this user segment.

META FIT

None natural.

WOW MOMENT

Modest. A chart and a sentence. The insight is profound and the visual forgettable, which is the wrong shape for a short demo.

FEASIBILITY

Fine on synthetic data, but forecast quality on four months of sparse history will look visibly shaky if a judge probes.

BIGGEST FLAW

One product decision away from being predatory. The moment the “you'll be short” nudge points at Paytm Postpaid, it stops being a user-impact feature and becomes a lending funnel — and a sharp judge will ask whether the roadmap ends there.

Paytm 18 · AI 15 · Impact 17 · Demo 12 · Feasibility 14 → 76

8. Chitra Mode — the literacy-free payment surface

MECHANISM

A mode where the interface is pictures and audio rather than text — but the differentiator is that the icons are *personal*, generated from the user's own history. The kirana shop appears as its actual storefront photo, mother as her contact photo, the electricity bill as a recognisable bulb card with the amount spoken aloud. Everything narrates in the user's language on focus.

WHY AI IS NECESSARY

Moderate. Personalised icon synthesis plus merchant categorisation plus text-to-speech. Each piece is available; the assembly is the idea.

WHY PAYTM IS NECESSARY

Moderate. It is an interface layer, and a judge will note that most of it is design thinking rather than machine learning.

META FIT

None natural.

WOW MOMENT

Reasonable, and the accessibility narrative is emotionally strong in a room evaluating AI for Paytm users. A demo where someone who cannot read completes a payment lands well.

FEASIBILITY

Medium. Building a genuinely parallel UI surface in one day is more work than teams estimate, and half-built alternate UIs look broken rather than minimal.

BIGGEST FLAW

Risks reading as a design project with AI sprinkled on. Under a rubric allocating a full 20 points to AI innovation, that is expensive.

Paytm 14 · AI 14 · Impact 17 · Demo 16 · Feasibility 13 → 74

Part I scoreboard

#	CONCEPT	PAYTM	AI	IMPACT	DEMO	FEASIBILITY	TOTAL
1	Rakshak	20	16	19	17	13	85
3	Ek Photo Setup	19	15	16	19	15	84
2	Bol-ke-Bhej	17	13	18	18	16	82
4	Paisa Kahan Gaya	18	14	15	13	17	77
5	Parchi	15	17	14	18	12	76
7	Nakad Nazar	18	15	17	12	14	76
8	Chitra Mode	14	14	17	16	13	74
6	Dispute Autopilot	19	12	18	11	12	72

The strongest two, defended

Rakshak wins on the one criterion that cannot be manufactured: it is the only concept here that is *structurally impossible* for a competitor to copy. Everything else could be built by a well-funded startup with API access; a mule-account graph spanning eleven states exists inside roughly three companies in India. It also targets trust, the theme's hardest word — speed and experience are engineering problems, trust is a data problem, and Paytm is one of the few players holding the data. The 13 on feasibility is the honest weakness, and the mitigation is aggressive transparency on stage: show the synthetic graph, show the feature schema, and state plainly that the model is trained on simulated data with a production path requiring no new instrumentation. Judges forgive a mocked dataset far more readily than a mocked idea.

Ek Photo Setup wins on the opposite axis: it is the most likely to actually work in the room, and it hits the brief's explicit mention of first-time users at precisely the moment they are lost. Onboarding is where Paytm bleeds regional and first-time users, and this does not smooth the form — it deletes the form. Nineteen on demo quality is not generous; a fresh install going from crumpled paper to a configured biller in under five seconds is the kind of thing a judge repeats to someone else afterwards, which is the real measure of a hackathon demo.

They are complementary rather than redundant, which matters if the team can split. Rakshak owns trust, Ek Photo Setup owns speed, and neither one's failure mode is the other's.

The one to argue against, despite its 82, is **Bol-ke-Bhej**. It will demo beautifully and it will lose, because three other teams will submit voice input and the judge scoring AI innovation has already seen speech-to-text. If the team insists, the entire pitch has to be the vernacular read-back as a fraud countermeasure — the moment it is framed as “payments, but faster to say,” it is indistinguishable from the pile.

Part II — Theme 2: AI for Small Businesses

Rejected before starting

These will be roughly 60% of the competing field, and they lose on AI innovation and impact:

- **AI dashboard for merchant insights.** Insight without an action is a report. Small merchants do not read reports; they run a shop with their hands full.
- **WhatsApp chatbot to answer customer questions.** The AI does the *easiest* part of the job, and there is no Paytm necessity — swap in any PSP and nothing changes.
- **Payment / udhaar reminder bot.** A cron job with a language model taped on. Judges will have seen it four times before lunch.
- **AI ad copy generator.** No Paytm surface touched, and the merchant's problem was never copywriting.
- **Voice-based analytics.** Demos beautifully for thirty seconds, then reveals itself as a text-to-SQL wrapper.

The filter applied to everything below: does the AI do something a rule engine provably cannot, does the concept die without Paytm's specific rails, and does money visibly move on stage?

1. Growth — “Kaunsa Ad Chala?” (closed-loop offline ad attribution)

MECHANISM

A Click-to-WhatsApp ad click carries Meta's click ID in the referral payload. The system mints a Paytm dynamic QR or payment link bound to that click, so when the customer walks in and pays, the Paytm webhook fires an offline Purchase event with real rupee value back to Meta's Conversions API. Ad optimisation now runs on in-store revenue instead of clicks. Walk-ins that never got a tagged QR are matched probabilistically.

WHY AI IS NECESSARY

The matching problem is the whole product: reconciling unattributed soundbox transactions against ad clickers using time windows, amount fingerprints, hashed phone numbers and first-message intent — then generating the next creative variant from what the converting cohort actually said.

WHY PAYTM IS NECESSARY

Paytm holds the only trustworthy record that a rupee arrived. The QR *is* the conversion event.

META FIT

Native and non-negotiable — CTWA click IDs, Conversions API, Advantage+ audiences.

WOW MOMENT

A judge clicks the ad on their own phone, pays ₹100 via the QR, and Events Manager shows an attributed ₹100 purchase seconds later. The agent then reallocates: “Ad B costs ₹31 per paying customer, Ad A costs ₹214.”

FEASIBILITY

Weak. Requires a live WABA, an approved ad account, and real (small) spend. Ad review latency alone can eat the sprint.

BIGGEST FLAW

Beyond the plumbing risk, small merchants spend too little for Meta's optimiser to ever learn. You would be selling attribution truth to someone spending ₹300 a day.

Paytm 18 · AI 17 · Impact 19 · Demo 18 · Feasibility 12 → 84

2. Growth — “Dhanda Meter” (demand-trough yield management)

MECHANISM

Reads per-15-minute Paytm transaction velocity per store, detects a demand trough forming against the same-weekday baseline, and mints a time-boxed offer *applied at the QR itself* — not a coupon code — broadcast to nearby opted-in customers on WhatsApp. The offer auto-cancels the moment velocity recovers. Airline yield management for a chai shop's dead 4–6pm window.

WHY AI IS NECESSARY

Forecasting a counterfactual for a single store with weeks of data needs hierarchical borrowing across similar merchants, plus a model that can encode local context a rule cannot (exam season, rain, a cricket match). Discount depth requires an elasticity estimate.

WHY PAYTM IS NECESSARY

Real-time transaction stream, offer applied at the Paytm QR, settlement confirms redemption.

META FIT

WhatsApp broadcast plus a three-hour local Advantage+ flight.

WOW MOMENT

The system does not just fire a discount — it *kills* the discount mid-flight when the shop gets busy. “The shop stops discounting itself the second it doesn't need to” is a line judges remember.

FEASIBILITY

Medium. The demo needs a synthetic transaction stream; discount-at-QR is the hard integration and would likely be faked as post-payment cashback.

BIGGEST FLAW

Tuning discount depth on one day of data is fiction, and merchants correctly fear training customers to wait for the 5pm price. The real bottleneck is that most Paytm merchants do not own their customers' phone numbers, so there is no list to broadcast to.

Paytm 17 · AI 18 · Impact 16 · Demo 17 · Feasibility 13 → 81

3. Money — “Paisa Timeline” (working-capital autopilot)

MECHANISM

Builds a 30-day cash calendar from Paytm settlement inflows plus scheduled outflows (vendor payouts, rent, salary, GST, EMI), finds the exact date the merchant goes negative, then picks the cheapest bridge and *executes* it: shift a specific vendor payment four days via an AI-drafted WhatsApp negotiation, accelerate a receivable, or draw the smallest possible slice of merchant credit for exactly N days.

WHY AI IS NECESSARY

Forecasting inflow from noisy daily UPI, classifying outflows from payout narrations, ranking remedies as a constrained optimisation, and writing a vendor message credible enough in the vendor's own language that he actually says yes.

WHY PAYTM IS NECESSARY

Settlement data is the ground truth of a small merchant's revenue, and Paytm payouts plus merchant lending are the levers. Very few players hold both the sensing and the acting side.

META FIT

WhatsApp Flows let the vendor accept a new date with one tap, writing straight back into the payout schedule. Negotiation genuinely lives on WhatsApp; this is not bolted on.

WOW MOMENT

“On the 14th you're ₹41,000 short. I asked your flour supplier to take payment on the 18th — he said yes — so you don't need a loan. That saved ₹1,900 in interest.” The vendor's reply visibly lifts the cash curve above zero on stage.

FEASIBILITY

Good. Settlement plus payout plus WhatsApp Flows; the forecast can be deliberately light.

BIGGEST FLAW

Half the outflows are not on Paytm — cash, other bank accounts — so the calendar is partially blind, and a wrong “you'll go negative” prediction destroys trust instantly. Nudging vendors also risks the merchant's most important relationships, and real loan-drawdown rails will not be available in a day.

Paytm 19 · AI 16 · Impact 20 · Demo 17 · Feasibility 15 → 87

4. Money — “Galla Guard” (revenue leakage at the counter)

MECHANISM

Reconciles what *should* have been collected against what actually reached Paytm, per staff shift. Targets the classic small-business leak: staff pocketing cash, or quietly telling customers to pay a personal UPI handle. The signal is the *absence* of Paytm where Paytm should have been — bills with no matching payment, a shift's average ticket drifting, regulars who stopped appearing. Output is a shift-level leak score with the three specific transactions to check.

WHY AI IS NECESSARY

Anomaly detection on sparse multi-modal signals where “expected collection” must be learned, not configured. The model also drafts an evidence-based, non-accusatory confrontation script for the owner.

WHY PAYTM IS NECESSARY

Soundbox events plus settlement are the only tamper-resistant ledger a shop like this has.

META FIT

Thin, and worth saying plainly: a WhatsApp shift-close summary to the owner and staff acknowledgement. This is the one concept where Meta is decoration.

WOW MOMENT

Plant a skim in the demo data and it says “between 7:12 and 7:40pm: nine bills, six payments, ₹1,340 missing — here are the three bills.” And it is right.

FEASIBILITY

Medium. It needs a billing signal to compare against; without one the accusation is unfalsifiable.

BIGGEST FLAW

The false-positive cost is social, not statistical — wrongly accusing a staff member is unrecoverable, and cash-heavy shops make expected collection genuinely unknowable. It is also a surveillance product, which some judges will react to viscerally.

Paytm 17 · AI 16 · Impact 17 · Demo 16 · Feasibility 13 → 79

5. Inventory — “Stock ka Andaza” (inventory inferred from payments)

MECHANISM

SKU-level stock tracking where the merchant never scans a barcode or counts anything. Two inputs: the histogram of Paytm transaction *amounts* (the amount is a basket fingerprint — ₹40 appearing 212 times means roughly 212 chai-samosa combos) and vendor purchase outflows via Paytm business payments (what he bought, from whom, how much). The AI deconvolves amounts into basket mix, maintains a probabilistic stock level per item, and fires reorder as a pre-filled Paytm payment plus a WhatsApp order to the supplier he already pays.

WHY AI IS NECESSARY

A real inverse problem — mixture decomposition plus Bayesian state estimation — with no rule-based path whatsoever. Supplier invoices arriving as WhatsApp photos feed unit costs back in via vision models.

WHY PAYTM IS NECESSARY

Both signals — the amount stream and the vendor payout ledger — exist only inside Paytm, and the reorder executes as a Paytm payment. Strip Paytm out and there is literally no input.

META FIT

Excellent and honest: supplier ordering already happens on WhatsApp, and the invoice photo returns through the same channel.

WOW MOMENT

“You'll run out of Thums Up on Thursday afternoon. Order 4 crates from Sharma Traders?” One tap sends the WhatsApp order and queues the payment. Inventory management with zero data entry, which every merchant has failed at before.

FEASIBILITY

The risky one. Full decomposition is not a one-day build; it becomes feasible only when scoped hard to a fixed 10–15 item menu where the amount-to-basket mapping is identifiable.

BIGGEST FLAW

Identifiability collapses under cash sales, overlapping prices and custom combos. A kirana with 2,000 SKUs breaks it outright, and a menu or price list is needed to bootstrap — so the cold-start story is uncomfortable.

Paytm 18 · AI 20 · Impact 18 · Demo 19 · Feasibility 13 → 88

6. Billing — “Khata Closer” (collections strategist, not a reminder)

MECHANISM

For each udhaar customer it learns personal repayment physics from Paytm khata plus historical inflow timing — this one pays on the 7th, this one only after being asked in person, this one responds to part-payment offers. It then chooses the instrument: a UPI collect request, a bounded three-installment autopay mandate, a pay-on-next-visit QR, or escalation to the owner for a face-to-face ask. It negotiates through a WhatsApp Flow, with acceptable settlement options constrained by the merchant's own cash position.

WHY AI IS NECESSARY

Per-person propensity from tiny samples, joint choice of channel, timing and ask-amount, and generating relationship-preserving language in the customer's dialect. A reminder cron is precisely the thing that already fails here.

WHY PAYTM IS NECESSARY

Business khata, UPI collect, autopay mandates for installments, and settlement callbacks that close the ledger automatically.

META FIT

WhatsApp Flows carry the negotiation and the payment in one thread.

WOW MOMENT

A judge plays the debtor, is offered three options, taps “₹500 now, ₹500 on the 5th,” a mandate is created, the ledger splits itself, and the merchant's cash position updates.

FEASIBILITY

The best on this list. This is the one a team could genuinely finish.

BIGGEST FLAW

It lives one framing mistake away from the reminder bot you were told to reject — if the pitch slips, AI innovation points vanish instantly. Collections templates also face WhatsApp approval risk and real harassment-conduct sensitivities.

Paytm 19 · AI 14 · Impact 19 · Demo 18 · Feasibility 18 → 88 (weakest single leg on the board)

7. Billing — “Bol ke Bill” (spoken order to exact-amount collection)

MECHANISM

The merchant speaks the order in natural Hinglish while still serving — “do chai, ek samosa, aur wo kal ka udhaar” — and the system builds the bill, then pushes the exact amount as a dynamic QR to the soundbox or terminal so the customer scans a pre-filled total. The payment confirmation closes the bill. It learns the merchant's private vocabulary: “chhota,” “half,” “family pack,” “usual.”

WHY AI IS NECESSARY

Streaming speech recognition in a noisy, code-mixed environment, plus entity resolution against a merchant-specific menu partly *learned from historical transaction amounts*. Deterministic parsing has no chance.

WHY PAYTM IS NECESSARY

The amount push and the transaction callback are the entire loop. Without them it is a calculator.

META FIT

Moderate at best — a WhatsApp receipt and reorder link, which does convert the customer into a retargetable contact. Do not oversell it.

WOW MOMENT

Four seconds, no typing, and the device announces “₹95 received.” The most physically impressive thing on this list in a live room.

FEASIBILITY

Speech and parsing are fine. Pushing an amount to a physical soundbox is the blocker — without the device or that API you would demo a dynamic QR on a phone screen, visibly weakening the story.

BIGGEST FLAW

Paytm integration depth is shallow unless amount-push on hardware is actually available, and fundamentally it saves seconds rather than making money. Judges over-reward the demo and under-notice that the impact ceiling is a time-saver.

Paytm 15 · AI 17 · Impact 16 · Demo 20 · Feasibility 16 → 84

8. Planning — “Agle Mahine ka Plan” (cohort-benchmarked expansion bets)

MECHANISM

Small merchants make roughly three real decisions: hire someone, extend hours or add a counter, add a product line. This benchmarks the merchant against anonymised cohort statistics from Paytm's own merchant base — same pincode, same category, same ticket size — to answer “what do shops like mine earn in the 9–11pm hour?” It then converts the chosen bet into a dated, funded plan: salary as a recurring payout, stock as a vendor payment, a small Meta flight to announce the change — plus a pre-committed kill criterion measured on Paytm transactions.

WHY AI IS NECESSARY

Cohort matching rather than naive averaging, translating a fuzzy ambition into a resourced plan, and setting a statistically defensible stopping rule.

WHY PAYTM IS NECESSARY

The single most “only Paytm could build this” idea here — millions of small-merchant transaction profiles keyed by pincode and category is a data asset no competitor can fake.

META FIT

The announcement ad and customer broadcast, with measurement closing back on Paytm transactions.

WOW MOMENT

“Shops like yours within 2km make ₹1,150 in the 9–11pm hour. You close at 9. Here's a 14-day test, it costs ₹6,800, and here's the exact number that means stop.” Pre-committing to a kill criterion is an unusually mature move in a hackathon.

FEASIBILITY

Poor. The cohort dataset will not be available, so it would be synthesised — and experienced judges smell synthetic benchmarks immediately.

BIGGEST FLAW

The entire value rests on inaccessible data, and the privacy story (k-anonymity or differential privacy over merchant revenue) is a compliance programme, not a one-day feature. Great slide, hollow build.

Paytm 20 · AI 15 · Impact 18 · Demo 15 · Feasibility 11 → 79

Part II scoreboard

CONCEPT	AREA	PAYTM	AI	IMPACT	DEMO	FEASIBILITY	TOTAL
Stock ka Andaza	Inventory	18	20	18	19	13	88
Khata Closer	Billing	19	14	19	18	18	88
Paisa Timeline	Money	19	16	20	17	15	87
Kaunsa Ad Chala?	Growth	18	17	19	18	12	84
Bol ke Bill	Billing	15	17	16	20	16	84
Dhanda Meter	Growth	17	18	16	17	13	81
Galla Guard	Money	17	16	17	16	13	79
Agle Mahine ka Plan	Planning	20	15	18	15	11	79

No concept clears 90 — the honest read on a one-day sprint, because feasibility is where every ambitious idea bleeds five to eight points.

The strongest two, defended

Primary: Stock ka Andaza. The only concept where the AI is doing something genuinely hard and genuinely unfakeable, and where removing Paytm removes the input entirely rather than just the payment step. It also solves the problem small merchants have failed at most consistently, because every prior attempt demanded data entry they will never do. The WhatsApp supplier loop is real behaviour, not a bolted-on Meta requirement. The condition: scope to a single vertical with a fixed, small menu — a tea stall, a QSR counter, a salon — and say so out loud, framing it as deliberate rather than hiding it. Show the probability distribution over stock, not a fake integer, and let a judge watch it sharpen as transactions arrive. If by mid-afternoon the decomposition is not holding on your own demo data, that is the signal to fall back.

Secondary: Paisa Timeline. The fallback and, for a judge who thinks like an operator, possibly the winner. Cash-flow timing is what actually kills small businesses, the impact leg scores a clean 20, and it is the concept where Paytm's sensing side and acting side are both indispensable. The vendor accepting a new date over WhatsApp and the cash curve lifting above zero is a complete story arc in fifteen seconds. Be upfront about the blind spot — Paytm inflows are clear, non-Paytm outflows barely visible — and turn it into the roadmap slide instead of letting a judge find it.

Two things to resist. Do not merge Paisa Timeline with Khata Closer just because receivables acceleration is technically a lever in the cash calendar; that is how one-day sprints produce two half-built demos. And do not pick Khata Closer as the headline despite its 88 — its score is carried by feasibility, its AI leg is the weakest on the board, and “we built smart payment reminders” is exactly the pitch you were told not to bring.

Part III — Finding the winning mechanism

The single strongest non-obvious mechanism

Use Paytm payment events as the supervision signal for a merchant's WhatsApp/Instagram selling agent.

The obvious framing is “put an AI chatbot in the merchant's DMs.” That is not the mechanism. The mechanism is the *join*: Meta owns the conversation, Paytm owns the outcome, and nobody currently connects a thread's state to whether money actually arrived. Once connected, every DM thread becomes a labelled training example — features (what was asked, what was quoted, how fast, what nudge was used) paired with ground truth (paid / not paid / how much / how long it took). That label is the scarce asset. Conversational AI is a commodity; a reward signal on Indian SMB selling behaviour is not, and Paytm is one of very few entities that can generate it at scale.

The product surface is a **Deal Ledger**, not an inbox: every inbound conversation is a deal with an explicit state (Enquiry → Priced → Link sent → Paid → Dead), the closing action is always a scoped, amount-locked, expiring Paytm payment link, and the Paytm webhook is what advances state. The headline metric is rupees recovered from threads that would have died.

This matters because the actual revenue leak for a small Indian merchant is not the abandoned cart — they have no cart. It is the thread that goes “kitna hai?” and then silence. That leak is invisible today precisely because conversation state and payment state live in different companies.

Why this beats the consumer-side field

ARCHETYPE	PAYTM INTEGRATION	AI INNOVATION	USER IMPACT	DEMO QUALITY	FEASIBILITY
Payment-supervised selling agent	Load-bearing	High	High, quantified	Ends with real money	One day
Scam / QR fraud detection	Load-bearing	Medium	High, unverifiable	Poor	Medium
Voice / vernacular UPI, accessibility	Medium	Low	High	Sympathetic but flat	High
Spend analytics / PFM copilot	Decorative	Low	Low	Flat	High
Merchant back-office (GST, inventory, recon)	Weak	Low	Medium	Flat	High

Consumer trust ideas fail on two criteria simultaneously, and the failures are structural rather than fixable. First, **prevention does not demo**. You cannot show a scam that did not happen; you have to stage a fake fraudster, and every judge recognises the theatre. Second, **you are competing with Paytm's own risk team**, who have the adversarial data you do not and are already funded to do this. The unspoken judge reaction to a fraud-detection pitch is “we'd build this internally anyway, and better.”

Accessibility and vernacular voice payments fail differently: UPI 123Pay, Bhashini and Paytm's existing voice surfaces have already commoditised the demo. Asked “what is the AI contribution beyond calling an ASR endpoint,” there is no good answer, and the innovation score collapses even though impact reads high.

The strongest consumer steelman is stranger-to-stranger transaction safety — making a marketplace deal between two people who do not trust each other survivable, given UPI's irreversibility. It is genuinely non-obvious and genuinely valuable. It loses anyway, because the only real fix is holding funds, Paytm cannot unilaterally escrow without regulatory work, and a one-day build reduces to a risk score with no ground truth. A good company and a bad hackathon project.

Merchant back-office ideas lose on drama and integration depth. A ledger that reconciles itself is useful and completely unwatchable, and most would work identically on any PSP, which guts the Paytm-integration score.

Why the Meta fit is natural rather than bolted on

Meta already monetises business messaging in India through Click-to-WhatsApp ads, and the funnel works up to the last step. Ad → DM → conversation → *and then nothing*, because there is no native close. The merchant types a price, the customer says “ok bhaiya,” and the deal evaporates. The missing primitive is a payment instrument living inside the thread, and in India that is UPI. So the seam is real: WhatsApp Business Cloud API and Instagram Messaging API supply demand and identity, Paytm supplies the close and the truth. This is not putting a Paytm logo on a Meta surface; it is completing a funnel Meta cannot complete alone.

What twenty other teams will build, and why they are weak

Roughly ten will be consumer: a support chatbot, spend analytics with LLM summaries, an “is this QR safe” scanner, fake-payment-screenshot detection, a bill-reminder agent, a loan-eligibility explainer, receipt OCR, a deals/shopping agent, an elderly-friendly voice mode, and a vernacular payment assistant. Roughly ten will be merchant: a WhatsApp order bot, catalogue generation from photos, voice inventory, a GST assistant, transaction-data credit scoring, Instagram creative generation, soundbox upsell, loyalty campaign blasts, natural-language business dashboards, and settlement reconciliation.

Their shared weakness is that they are all **screens wrapped around a model call**. Three failure modes recur:

- **Removable.** Swap Paytm for Razorpay or Stripe and the demo is unchanged — fatal on integration.
- **Unsupervised.** Nothing in the design improves with usage, so the answer to “what happens after month one” is “the same thing.”
- **Undemoable.** The peak moment is a chat bubble appearing, which after the fourth team produces no reaction at all.

There is also a timing problem nobody accounts for: at least four teams will build the same WhatsApp order bot and be judged against each other in sequence. Being the fifth chat window of the afternoon is a scoring catastrophe independent of quality.

Red-teaming the pick

“This is just Wati/AiSensy/Interakt with a Paytm link.”

The most dangerous objection, and partly fair. The defence is that those tools sell an inbox and charge for message volume; they have no reason to optimise for whether money arrived, and no access to the signal. But this defence only holds if you never show a chat UI as the product. Show the ledger and the recovered-revenue number. If a judge remembers the demo as “a chatbot,” you have lost regardless of what you built.

Meta's messaging policy is the sharpest constraint.

WhatsApp Business Cloud API confines free-form replies to a 24-hour customer service window; re-engaging outside it requires pre-approved templates, marketing templates cost money, and Meta paces per-user marketing delivery and enforces opt-outs. So “automatically chase every dead thread at hour 40” is not straightforwardly legal on the platform. This reshapes the design rather than killing it: most value sits inside the 24-hour window, where speed-to-quote is the dominant variable, and out-of-window chases go through a small set of pre-approved utility templates. Start template approval days before the event — it is the only long-pole item and it is not under your control on the day.

Auto-nudging is a self-poisoning strategy.

Aggressive follow-ups tank the merchant's WhatsApp quality rating, and Meta throttles low-rated senders. A naive conversion-maximising policy will destroy the channel it depends on. The right framing — a genuine sophistication point in front of judges — is a per-customer nudge budget with the policy optimising contribution margin net of quality-rating cost, not raw conversions. Show the system *declining* to chase someone.

The learning loop cannot actually train in one day.

You will have no data. Do not fake a model; judges who catch a fabricated learning curve stop believing everything else. Ship a cold-start heuristic, show the logging schema explicitly, and present it as bandit-ready with the reward already defined. “We have the label and the reward, we don't yet have the volume” is a credible, adult answer.

Extraction errors cost the merchant real money.

A misread quantity or price means a refund and a lost customer. Keep a human-in-the-loop approve tap for every quote. This is also the right demo choice, because it converts a failure mode into a visible trust feature.

Reconciliation is messier than the happy path.

Customers pay twice, pay the wrong amount, or pay in cash and never touch the link. You need one link per deal with the amount locked, idempotent webhook handling, and a manual “closed outside the app” state. Say this out loud before a judge finds it.

Cold-start pricing.

The agent has to know prices, and these merchants have no catalogue. The non-obvious bootstrap is Paytm's own transaction history — the merchant's modal ticket sizes reveal most of the price list before they type anything. Worth surfacing explicitly, because it is another thing only Paytm can do.

“Paytm already has payment links and Paytm for Business.”

Correct, and that is the point: the primitive exists, the decision layer above it does not. Integration is a day's work precisely because the rails are there.

A deterministic three-minute demo

Design principle: exactly two things are live, everything else is a fixture, and the live things are ones you control. Pre-seed a demo merchant with about forty realistic historical threads, twelve of them dead, loaded from a fixture file with fixed seeds. No model call is needed to render the board.

TIME	BEAT	WHAT HAPPENS
0:00– 0:20	The leak	A real phone showing a real-looking WhatsApp inbox with a dozen unanswered “price?” messages. One sentence: this merchant did ₹X last month and lost ₹Y in threads that simply died.
0:20– 1:00	The ledger	Flip to the Deal Ledger. Every thread has a state the merchant never typed. A single recoverable-rupees number at the top. Do not narrate features; let the states speak.
1:00– 1:50	One live thread	A judge (or rehearsed second phone) sends a messy Hinglish voice note with a photo: “bhaiya ye wala 2 chahiye, Noida delivery.” The agent extracts item, quantity and address, drafts an itemised quote, the merchant taps Approve, and a Paytm link goes out. The only live model call in the entire demo. Pre-warm it, cache responses keyed by input hash so rehearsed inputs replay instantly, and keep a hardcoded fallback on timeout.
1:50– 2:20	The close	The judge pays ₹1 on the real link from their own phone. The board flips to Paid with an audio cue. This is the peak, and it is real money moving — which, after twenty chat-bubble demos, is the only thing anyone will remember.
2:20– 2:50	The loop	Show the row the payment just wrote: thread features, nudge used, quote latency, outcome. Then the policy panel: for this merchant, an itemised link within four minutes converts materially better than a bare price reply. Then show the nudge budget refusing to chase one customer to protect the quality rating. This thirty seconds is where the innovation score is won.
2:50– 3:00	Scale	Every Paytm merchant with a WhatsApp number is addressable on day one, and the training signal belongs to Paytm alone.

Guardrails to rehearse: a keyboard kill-switch that plays a recorded happy path with narration timed to match; a second SIM in case judges will not scan; frozen model version and seeds; cached responses for offline operation if venue wifi dies; WhatsApp template approval submitted well in advance. Assume the network fails and build the demo so that nobody can tell.

Part IV — Reconciling the three analyses

The three analyses were run independently and do not agree. That disagreement is the most useful thing in this dossier, so it is worth stating plainly rather than smoothing over.

Where they agree

- **Theme 2 is the stronger theme.** The merchant analysis produced three concepts at 87–88 against the consumer analysis's ceiling of 85, and the mechanism analysis argues the consumer side is structurally handicapped on demo quality and competitive positioning.
- **Nothing deserves 90+ in a one-day sprint.** Feasibility is where every ambitious concept bleeds points, and all three analyses scored it harshly.
- **The obvious ideas are unanimously rejected:** chat assistants, transaction-history Q&A, voice navigation, insight dashboards, reminder bots, ad-copy generators.
- **“Removability” is the decisive test.** If swapping Paytm for another PSP leaves the demo unchanged, the concept cannot score well on a rubric where Paytm integration is a full fifth of the marks.

Where they conflict

On consumer fraud prevention. Part I ranks Rakshak first overall at 85, arguing its moat is structurally uncopyable. Part III argues the same category is a trap: prevention cannot be demonstrated because you cannot show a scam that did not happen, and the team would be competing against Paytm's own funded risk organisation. Both are right about different criteria — Rakshak is the best *product* on the consumer list and among the worst *hackathon demos*. Part I's own feasibility score of 13 concedes most of this.

On what carries a demo. Part I rewards emotional and visual surprise (a crumpled bill becoming a configured biller; an amber warning in Tamil). Part III argues that in a room of fifty demos only *verified money movement* survives as a memory, which systematically favours merchant concepts that end in a real payment.

On the learning loop. Part III treats an outcome-supervised loop as the core innovation claim while conceding it cannot actually be trained in a day. Part II's top pick, Stock ka Andaza, makes a harder technical claim (mixture decomposition over payment amounts) that is demonstrable immediately but breaks under cash sales and overlapping prices.

The consolidated read

QUESTION	ANSWER ACROSS THE THREE ANALYSES
Strongest overall	Payment-supervised selling agent / Deal Ledger (Part III), which is Part II's growth-and-billing insight generalised into a mechanism
Safest to finish	Khata Closer (Part II, feasibility 18) or Ek Photo Setup (Part I, feasibility 15)
Most technically innovative	Stock ka Andaza (Part II, AI 20)
Best live demo	Bol ke Bill (Part II, demo 20); any concept ending in a judge paying real money
Best business	Paisa Timeline (Part II, impact 20) — cash-flow timing is what actually kills small businesses
Most defensible moat	Rakshak (Part I) on data; Deal Ledger (Part III) on the outcome label
Highest-risk trap	Parchi (handwriting accuracy on stage) and Agle Mahine ka Plan (synthetic cohort benchmarks)

The common thread across all three analyses is a single filter worth carrying into the build: **choose a narrow, failure-prone Paytm workflow; ground the AI in verifiable events; require explicit human confirmation before anything commits; and close the loop with a real Paytm action whose result is measurable on stage.**

Paytm AI Hackathon — Concept Research Dossier. All scores are independent analyst estimates produced for internal decision-making and are not official hackathon marks. Concepts referencing Paytm internal datasets (mule-account graphs, cohort merchant benchmarks, settlement history) assume synthetic stand-in data for any prototype; production paths are noted per concept.